

PALMER, AK, USA

WMO: 702740

Lat: 61.596N

Lon: 149.092W

Elev: 230

StdP: 14.57

Time Zone: -9.00 (NAK)

Period: 97-19

WBAN: 25331

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-8.5	-20.1	1.8	-10.3	-15.3	2.4	-2.9	41.3	21.8	36.6	25.3	1.5	0	0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	75.0	60.4	71.7	58.4	68.4	56.8	61.8	73.4	59.8	69.6	58.0	66.4	7.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.3	68.1	65.7	54.7	64.2	63.7	53.5	61.5	62.3	27.6	73.5	26.3	69.6	25.1	66.5	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.1	25.0	20.1	DB	-16.6	78.9	7.9	4.0	-22.2	81.7	-26.8	84.1	-31.2	86.3	-37.0	89.2	(4)
			WB	-16.8	64.2	7.6	2.2	-22.3	65.8	-26.7	67.0	-31.0	68.2	-36.5	69.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	16.5	21.4	26.3	38.6	49.0	56.1	59.0	56.3	48.6	36.5	22.6	18.4
	DBStd	17.81	14.64	11.99	9.88	6.70	5.40	4.19	3.68	3.91	4.96	8.59	11.82	12.67
	HDD50	5305	1037	801	734	345	83	2	0	2	79	422	821	979
	HDD65	10030	1502	1221	1199	793	496	269	188	271	492	884	1271	1444
	CDD50	755	0	0	0	2	52	185	280	196	37	3	0	0
	CDD65	6	0	0	0	0	0	2	3	1	0	0	0	0
	CDH74	131	0	0	0	0	5	38	64	24	0	0	0	0
	CDH80	11	0	0	0	0	0	4	6	1	0	0	0	0
Wind	WSAvg	6.7	6.6	6.9	7.7	8.3	8.1	6.5	5.6	5.4	6.1	6.2	6.5	7.0
Precipitation	PrecAvg	16.3	1.0	0.9	0.6	0.4	0.7	1.1	2.0	2.6	3.0	1.5	1.1	1.4
	PrecMax	21.7	2.6	2.7	2.1	1.4	2.4	2.6	4.2	6.2	5.8	3.0	2.5	3.6
	PrecMin	11.9	0.1	0.1	0.0	0.0	0.0	0.0	0.4	0.8	1.0	0.5	0.1	0.0
	PrecStd	2.6	0.6	0.6	0.5	0.3	0.6	0.7	1.0	1.3	1.5	0.7	0.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	46.5	49.0	62.8	74.1	78.8	79.8	77.0	64.8	58.9	48.4	46.1
		MCWB	39.6	39.5	40.9	48.8	55.4	61.4	62.7	62.0	53.3	48.9	41.6	39.4
	2%	DB	43.0	41.9	44.8	56.0	68.7	73.4	75.3	72.5	61.5	53.2	45.0	42.9
		MCWB	37.6	36.3	37.4	43.7	52.4	58.8	61.6	59.6	52.1	45.3	39.6	37.7
	5%	DB	39.3	39.2	41.9	52.4	64.2	70.0	71.8	68.2	59.1	50.1	41.3	38.8
		MCWB	35.0	34.9	35.6	41.6	50.5	57.3	59.8	57.8	50.9	43.5	36.8	34.7
	10%	DB	36.0	36.6	39.1	49.7	60.7	66.2	68.4	64.8	56.8	47.6	37.8	35.5
		MCWB	32.2	32.6	33.8	40.3	48.4	55.0	57.9	55.8	49.7	42.1	34.2	31.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.2	40.2	41.4	49.3	56.4	63.0	64.4	63.5	55.2	49.7	42.7	40.2
		MCDB	45.8	45.7	48.2	62.1	72.1	75.8	77.0	74.5	61.9	57.5	47.0	44.8
	2%	WB	37.9	37.1	38.4	44.9	53.2	60.2	62.3	60.7	53.5	46.5	39.8	37.9
		MCDB	42.4	41.0	43.3	53.7	66.6	71.7	74.0	70.7	59.6	51.6	44.3	42.4
	5%	WB	35.4	35.1	36.3	42.5	51.0	57.8	60.3	58.8	52.0	44.4	37.3	35.0
		MCDB	39.3	38.8	41.0	51.0	62.6	68.1	70.5	66.8	57.3	48.9	40.8	38.4
	10%	WB	32.2	32.9	34.3	40.8	49.2	55.7	58.7	56.8	50.7	42.6	34.6	31.9
		MCDB	35.6	36.5	38.8	48.7	59.6	64.9	67.2	63.5	55.5	47.0	37.6	34.8
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.8	14.6	16.5	18.6	16.9	15.6	15.0	13.8	12.0	11.1	11.6
		MCDBR	12.0	12.4	14.2	21.6	25.8	24.4	23.4	21.7	16.8	13.1	11.5	13.9
		MCWBR	8.8	9.5	9.9	12.4	13.2	12.7	12.2	12.0	9.7	8.2	8.4	10.9
	5% WB	MCDBR	11.7	11.6	12.9	19.9	23.5	22.2	21.2	19.3	14.8	11.6	11.1	12.8
	MCWBR	8.9	9.3	9.3	12.0	12.3	12.2	11.7	11.4	9.1	7.7	8.3	10.5	
Clear-Sky Solar Irradiance	taub	0.229	0.286	0.302	0.342	0.364	0.370	0.374	0.359	0.313	0.288	0.241	N/A	
	taud	2.071	2.151	2.194	2.152	2.195	2.312	2.344	2.396	2.493	2.336	2.035	N/A	
	Ebn at Noon	153	213	256	267	270	271	266	261	256	216	137	N/A	
	Edh at Noon	12	22	30	38	40	36	35	30	23	18	12	N/A	
All-Sky Solar Radiation	RadAvg	88	312	829	1388	1654	1690	1508	1197	776	357	129	45	
	RadStd	12	35	75	108	127	108	114	119	97	39	18	5	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	+3.56	+2.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page